



Barker College

Barker Student Number: _____

Mathematics Extension 1

2010
TRIAL
HIGHER SCHOOL
CERTIFICATE
PM FRIDAY 13 AUGUST

Staff Involved:

- GDH*
- LJP*
- BHC
- WMD
- TRW
- GIC
- VAB

85 copies

General Instructions

- **Reading time – 5 minutes**
- **Working time – 2 hours**
- **Write using blue or black pen**
- **Make sure your Barker Student Number is on ALL pages**
- **Board-approved calculators may be used**
- **A table of standard integrals is provided on page 13**
- **ALL necessary working should be shown in every question**
- **Marks may be deducted for careless or badly arranged working**

Total marks – 84

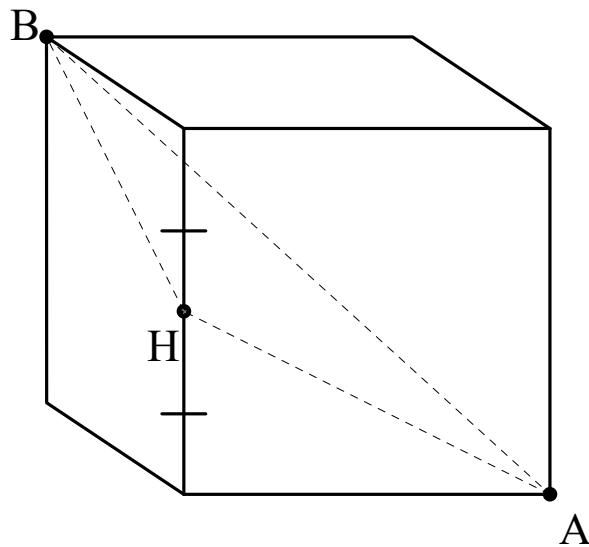
- **Attempt Questions 1 – 7**
- **All questions are of equal value**

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Question 1 (12 marks) [BEGIN A NEW PAGE]
Marks

(a) Solve $\frac{x-3}{x-2} < 1$ 2

- (b) The diagram shows a cube with edge length 2 units.
 H is the midpoint of the edge shown.
 Using triangle AHB or otherwise, find the size of $\angle AHB$ 3



(c) (i) Write as a single fraction: $\frac{1}{x+h} - \frac{1}{x}$ 1

(ii) Using $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$,

find $f'(x)$ from first principles if $f(x) = \frac{1}{x}$ 2

(d) Find $\int_{-3}^0 x\sqrt{1-x} \, dx$ using the substitution $u = 1-x$ 4

Question 2 (12 marks) [BEGIN A NEW PAGE]
Marks

- (a) The region under the curve $y = \cos 3x$ between $x = \frac{\pi}{18}$ and $x = \frac{\pi}{9}$ is rotated about the x -axis. **4**

Evaluate the volume of the solid of revolution formed.

- (b) Consider $f(x) = x^6 - 2x^4 + x^2$
- (i) Explain why $f(x)$ is an even function **1**
- (ii) Factorise $x^4 - 2x^2 + 1$ **1**
- (iii) Find the three x -intercepts of $f(x)$ **1**

For the remainder of the question, use part (i) to save time.

- (iv) Find the coordinates of the five stationary points of $f(x)$ **3**
- (v) Determine the nature of the stationary points **1**
- (vi) Sketch the graph of $y = f(x)$. Do not find points of inflexion. **1**

Question 3 (12 marks) [BEGIN A NEW PAGE]
Marks

(a) (i) Sketch $y = (\cos^{-1} x) - \frac{\pi}{2}$ showing all important features **2**

(ii) Differentiate $y = x \cos^{-1} x - \sqrt{1-x^2}$ **2**

(iii) Hence or otherwise evaluate $\int_{-1}^1 (\cos^{-1} x) - \frac{\pi}{2} dx$ **2**

(b) The acceleration of a particle moving along a straight path is given by

$$\ddot{x} = (\ln 3)^2 x$$

Initially, $x = 1$ and $v = \ln 3$

(i) Show that $v = (\ln 3)x$ **3**

(ii) Find x as a function of t **3**

Question 4 (12 marks) [BEGIN A NEW PAGE]
Marks

(a) Consider $f(x) = 2^{-x} + 1$

(i) Draw a large sketch $y = f(x)$ showing all important features **1**

(ii) Find $f^{-1}(x)$ **2**

(iii) Using parts (i) or (ii),
accurately sketch $y = f^{-1}(x)$ on the same set of axes as used in part (i) **1**

(b) A particle is moving in a straight line and its position x is given by the equation
 $x = \cos\left(3t - \frac{\pi}{2}\right)$ where x is the displacement in metres and t is the time in seconds.

(i) Prove that the particle is moving in Simple Harmonic Motion **2**

(ii) Sketch the displacement of the particle for the domain $0 \leq t \leq \pi$
showing all important features **2**

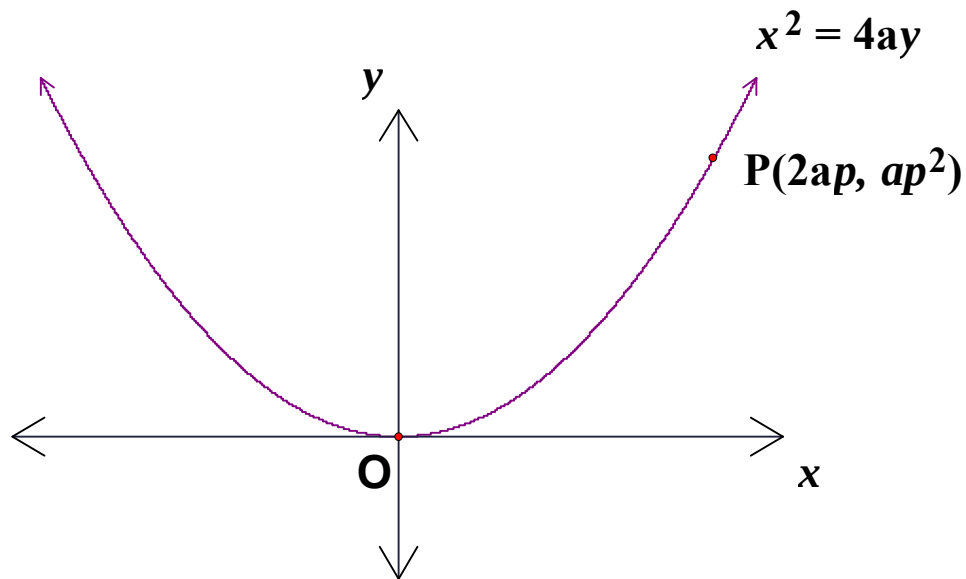
(iii) How far did the particle travel in the first π seconds? **1**

(iv) The displacement is currently defined in terms of cosine.
Define the displacement in terms of sine. **1**

(v) When is the first time that the particle is travelling at half its maximum speed? **2**

Question 5 (12 marks) **[BEGIN A NEW PAGE]**
Marks

(a)



- | | | |
|-------|---|----------|
| (i) | Find the gradient of the line perpendicular to OP | 2 |
| (ii) | The point A is on the x-axis. AP is parallel to the y-axis.
State the coordinates of the point A | 1 |
| (iii) | Find the equation of the line through A perpendicular to OP | 1 |
| (iv) | Through what point does the line in (iii) always pass? | 1 |

Question 5 continues on the next page

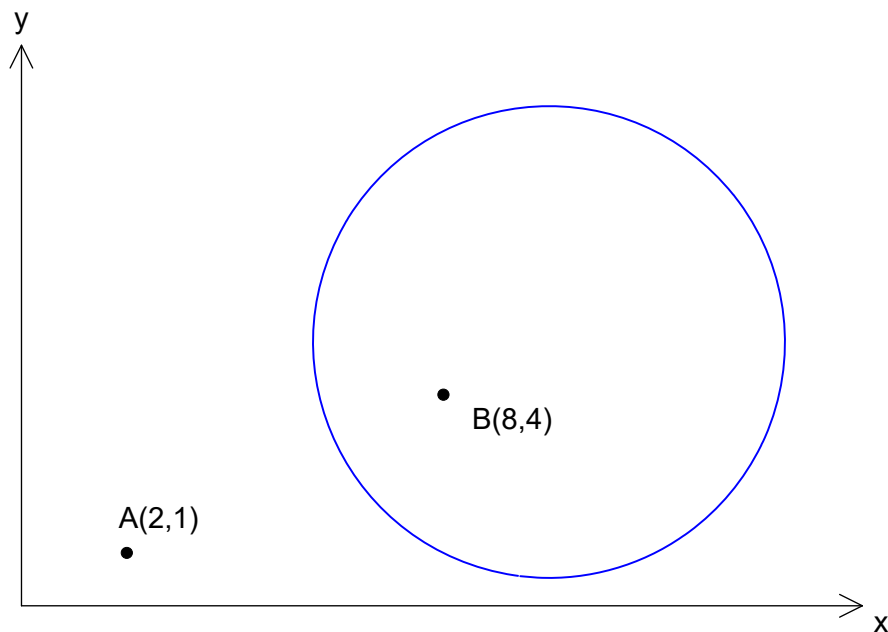
Question 5 (continued)
Marks

(b) Consider the points $A(2,1)$ and $B(8,4)$

- (i) The point I divides the interval AB **internally** in the ratio $2:1$.
Find the coordinates of the point I **1**
- (ii) The point E divides the interval AB **externally** in the ratio $2:1$.
Find the coordinates of the point E **2**

A point P moves such that it is always twice as far from $A(2,1)$ as it is from $B(8,4)$.

The locus of P is the circle as shown below.



- (iii) Using parts (i) and (ii) or otherwise, find the equation of the locus of P .
Your equation must be in the form where the centre and radius are easily determined. **4**

Question 6 (12 marks) [BEGIN A NEW PAGE]
Marks

- (a) (i) How many six letter 'words' can be made from the letters of the word
 CHANCE? 1
- (ii) What is the probability that one of the words in part (i) contains the word
 EACH? 2
- (iii) How many three letter 'words' can be made from the letters of the word
 CHANGE? 1
- (iv) How many three letter 'words' can be made from the letters of the word
 CHANCE? 2
- (b) Find the simplified sixth term of the binomial expansion of $\left(\frac{2x}{3} - \frac{3}{2x}\right)^{11}$ 2
- (c) Prove, by mathematical induction, the formula for the sum of an arithmetic series.
 i.e. Prove, by mathematical induction, that
- $$a + [a + d] + [a + 2d] + \dots + [a + (n-1)d] = \frac{n}{2} [2a + (n-1)d] \quad \text{for all integers } n \geq 1$$
- 4

Question 7 (12 marks) [BEGIN A NEW PAGE]

Marks

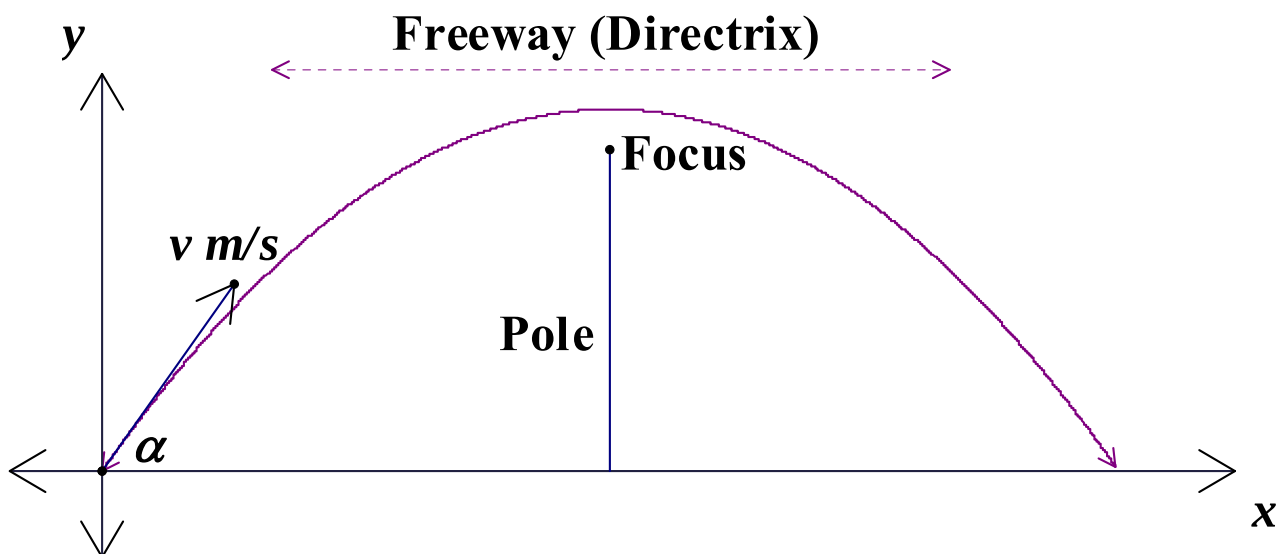
- (a) A particle moves on a curve with equation $y = 4 + 2 \tan^{-1}(3x^2 + 5)$.

$\frac{dx}{dt}$ is a finite constant.

What is the limiting value of $\frac{dy}{dt}$ as $x \rightarrow \infty$?

2

- (b)



The parabola frog (*Frogus projectilius*) jumps with initial velocity v m/s at an angle of projection α and its path traces a parabolic arc as shown above.

The frog's horizontal displacement from the origin, t seconds after jumping is given by the equation $x = vt \cos \alpha$ [Do not prove this].

The frog's vertical displacement from the origin, t seconds after jumping is given by the equation $y = vt \sin \alpha - \frac{1}{2}gt^2$ [Do not prove this].

- (i) Show that the frog lands after $\frac{2v \sin \alpha}{g}$ seconds

1

- (ii) Show that the frog's range is $\frac{v^2 \sin 2\alpha}{g}$ metres

1

Question 7 continues on the next page

Question 7 (continued)**Marks**

(iii) Show that the frog's maximum height is $\frac{v^2 \sin^2 \alpha}{2g}$ metres **2**

(iv) Show that the ratio of the frog's maximum height to range is $\frac{\tan \alpha}{4}$ **2**

(v) Let $\alpha_{\max}$ be the angle the frog jumps at to ensure maximum range.
Find the ratio of the frog's maximum height to its range in this case. **1**

(vi) Let $g = 10 \text{ m/s}^2$ and $v = \sqrt{85} \text{ m/s}$.

Let α_{equal} be the angle where the frog's maximum height equals its range.

At this angle, the frog needs a 40cm gap to squeeze between a pole and a freeway.

The top of the pole is the focus of the parabola and the freeway is the directrix.

Can the frog squeeze through? **3**

End of Paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

$$(1) (a) \frac{x-3}{x-2} - \frac{x-2}{x-2} < 0$$

$$\frac{x-3-x+2}{x-2} < 0$$

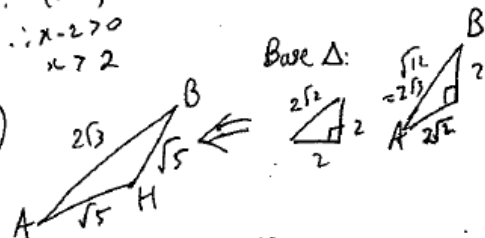
$$\frac{-1}{x-2} < 0$$

$\therefore -(x-2) < 0$ since if $\frac{a}{b} < 0$, $ab < 0$

$$\therefore x-2 > 0$$

$$x > 2$$

(b)



$$\cos \angle AHB = \frac{5+5-12}{2 \cdot 5 \cdot \sqrt{5}}$$

$$\cos \angle AHB = -\frac{1}{5}$$

$$\therefore \angle AHB = 101^\circ 32'$$

(c)

$$(i) \frac{x-(x+h)}{x(x+h)} = \frac{-h}{x(x+h)}$$

$$(ii) f'(x) = \lim_{h \rightarrow 0} \frac{\frac{x}{x+h} - \frac{x}{x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{-h}{x(x+h)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-1}{x(x+h)}$$

$$= \lim_{h \rightarrow 0} \frac{-1}{x^2}$$

$$= -\frac{1}{x^2}$$

$$(d) u=1-x \rightarrow x=1-u$$

$$du = -dx$$

$$x = -3, u = 4$$

$$x = 0, u = 1$$

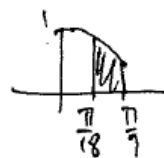
$$\therefore \int_4^1 (1-u) \sqrt{u} du$$

$$= \int_1^4 \sqrt{u} - u^{\frac{3}{2}} du$$

$$\text{or } (x-3)(x-1) < (x-2)^2 \quad (2) (a)$$

$$(x-2)[x-3-(x-1)] < 0$$

$$(x-2)(-1) < 0$$



$$V = \pi \int_{\pi/18}^{\pi/9} \cos^2 3x dx$$

$$\cos 2x = 2\cos^2 x - 1$$

$$\therefore \cos^2 x = \frac{\cos 2x + 1}{2}$$

$$\therefore \cos^2 3x = \frac{\cos 6x + 1}{2}$$

$$\therefore V = \frac{\pi}{2} \int_{\pi/18}^{\pi/9} \cos 6x + 1 dx$$

$$= \frac{\pi}{2} \left[\frac{\sin 6x}{6} + x \right]_{\pi/18}^{\pi/9} = \frac{\pi}{2} \left[\frac{\sin \frac{2\pi}{3}}{6} + \frac{\pi}{9} - \frac{\sin \frac{\pi}{3}}{6} - \frac{\pi}{18} \right]$$

$$= \frac{\pi}{2} \left[\frac{\pi}{18} \right] = \frac{\pi^2}{36}$$

$$(b) (i) f(x) = a^6 - 2x^4 + a^2$$

$$f(-a) = (-a)^6 - 2(-a)^4 + (-a)^2 = a^6 - 2a^4 + a^2 = f(a) \therefore f(x) = f(-x) \text{ - Even f'n}$$

$$(ii) (x^2)^2 - 2(x^2) + 1 = (x^2 - 1)^2$$

$$(iii) y=0 \therefore 0 = x^6 - 2x^4 + x^2$$

$$0 = x^2(x^4 - 2x^2 + 1)$$

$$0 = x^2(x^2 - 1)^2$$

$$\therefore x = 0, \pm 1$$

$$(iv) f'(x) = 6x^5 - 8x^3 + 2x = 0 \text{ - Set pt}$$

$$2x(3x^4 - 4x^2 + 1) = 0$$

$$2x(3x^2 - 1)(x^2 - 1) = 0$$

$$\therefore x = 0, \pm 1, \pm \frac{1}{\sqrt{3}}$$

$$\text{Set pts } (0,0)$$

$$(1,0) \quad (-1,0)$$

$$= \left(\frac{2}{3}u^{\frac{3}{2}} - \frac{2}{5}u^{\frac{5}{2}} \right)_1^4$$

$$= \frac{2}{3}(8) - \frac{2}{5}(32) - \frac{2}{3} + \frac{2}{5}$$

$$= -\frac{11}{15}$$

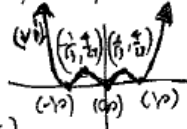
$$\left(\frac{1}{\sqrt{3}}, \frac{4}{27} \right) \quad \left(-\frac{1}{\sqrt{3}}, \frac{4}{27} \right)$$

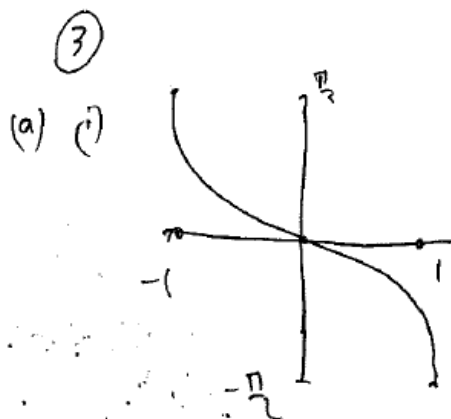
$$(v) f''(x) = 30x^4 - 24x^2 + 2$$

$$f''(0) = 2 \therefore \text{Min TP } (0,0)$$

$$f''(1) = 8 \therefore \text{Min TP } (1,0), (-1,0)$$

$$f''\left(\frac{1}{\sqrt{3}}\right) = -\frac{2}{3} \therefore \text{Max TP } \left(\frac{1}{\sqrt{3}}, \frac{4}{27}\right), \left(-\frac{1}{\sqrt{3}}, \frac{4}{27}\right)$$





(ii) $y = \cos^{-1}x - \frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{1-x^2}}$

$$= \cos^{-1}x - \frac{x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}}$$

$$= \cos^{-1}x$$

(iii) = 0 since odd fn (see graph)

from odd fn:

$$f(a) = \cos^{-1}a - \frac{\pi}{2}$$

$$f(-a) = \cos^{-1}(-a) - \frac{\pi}{2}$$

$$= \pi - \cos^{-1}a - \frac{\pi}{2}$$

$$= \frac{\pi}{2} - \cos^{-1}a$$

$$= -f(a)$$

or:

$$\left[x \cos^{-1}x - \sqrt{1-x^2} - \frac{\pi}{2}x \right]_{-1}^1$$

$$= \cos^{-1}1 - 0 - \frac{\pi}{2} + \cos^{-1}(-1) + 0 - \frac{\pi}{2}$$

$$= -\pi + \cos^{-1}(-1) = -\pi + \pi = 0$$

(b) $f=0, x=1, v=\ln 3$

(i) $\frac{d}{dt} \frac{1}{2}v^2 = (\ln 3)^2 x$

$$\therefore \frac{1}{2}v^2 = \frac{(\ln 3)^2 x^2}{2} + C$$

$$v^2 = (\ln 3)^2 x^2 + C$$

Now $x=1, v=\ln 3$

$$\therefore (\ln 3)^2 = (\ln 3)^2 + C \therefore C=0$$

$\therefore v^2 = (\ln 3)^2 x^2$ but when $x=1, v=\ln 3$

$$\therefore v = \pm (\ln 3)x$$

(ii) $\frac{dv}{dt} = (\ln 3)x$

$$\frac{dv}{dx} = \frac{1}{(\ln 3)x}$$

$$v = \frac{1}{\ln 3} \ln x + C$$

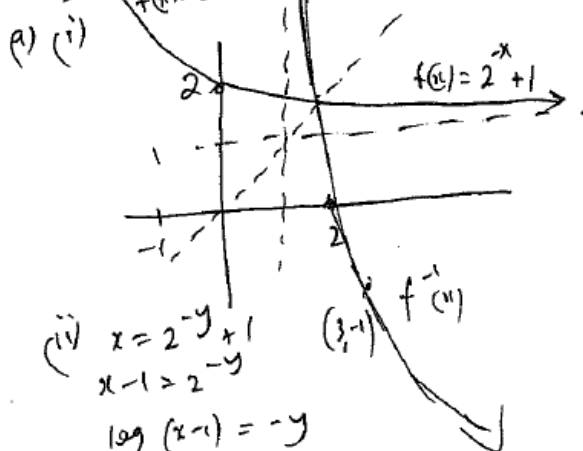
$x=1, v=0$

$$\therefore 0 = 0 + C \therefore C=0$$

$$\therefore v = \frac{\ln x}{\ln 3}$$

$t = \log_3 x$

$\therefore 3^t = x$



(ii) $x = 2^{-y} + 1$

$$x-1 = 2^{-y}$$

$$\log_2(x-1) = -y$$

$$\therefore f^{-1}(x) = -\log_2(x-1)$$

(iii) See above

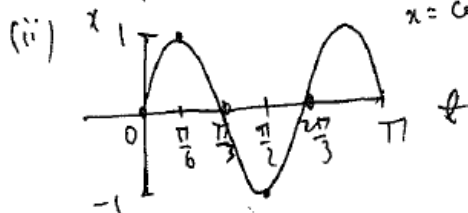
(b) (i) $x = -3 \sin(3t - \frac{\pi}{2})$

$$x = -9 \cos(3t - \frac{\pi}{2})$$

$$= -9 \sin 3t$$

$\therefore \sin 3t = \frac{x}{9}$ $n=3$

$$x = \cos(3(t - \frac{\pi}{6}))$$



(iii) 6 m

(iv) $x = \sin 3t$

(v) $v = 3 \cos 3t$

Max speed = 3 m/s

When is speed 1.5 m/s?

$1.5 = 3 \cos 3t$

$$0.5 = \cos 3t$$

$$3t = \frac{\pi}{3}$$

$$t = \frac{\pi}{9} \text{ sec}$$

$$(5) (a) (i) M_{OP} = \frac{ap^2}{2ap} = \frac{p}{2}$$

$$\therefore m_{perp OP} = -\frac{2}{p}$$

$$(ii) A(2ap, 0)$$

$$(iii) y-0 = -\frac{2}{p}(x-2ap)$$

$$y = -\frac{2}{p}x + 4a$$

$$(iv) (0, 4a)$$

$$(b) (i) (2, 1) \quad (8, 4)$$

$$2:1$$

$$I\left(\frac{2+8}{3}, \frac{1+4}{3}\right) = I(6, 3)$$

$$(ii) 2:-1$$

$$E\left(-\frac{2+16}{1}, -\frac{1+8}{1}\right) = E(14, 7)$$

$$(iii) \text{Diameter } (6, 3) \text{ to } (14, 7)$$

$$\therefore \text{Centre } (10, 5)$$

$$\text{Diameter} = \sqrt{8^2 + 4^2} = \sqrt{64+16}$$

$$= \sqrt{80}$$

$$= 4\sqrt{5}$$

$$\therefore \text{Radius} = 2\sqrt{5}$$

$$\therefore \text{eqn: } (x-10)^2 + (y-5)^2 = (2\sqrt{5})^2$$

$$(x-10)^2 + (y-5)^2 = 20$$

$$\text{or: } 2\sqrt{(x-8)^2 + (y-4)^2} = \sqrt{(x-2)^2 + (y-1)^2}$$

$$4[x^2 - 16x + 64 + y^2 - 8y + 16] = x^2 - 4x + 4 + y^2 - 2y + 1$$

$$3x^2 - 60x + 3y^2 - 30y = -315$$

$$x^2 - 20x + y^2 - 10y = -105$$

$$x^2 - 20x + 100 + y^2 - 10y + 25 = 20$$

$$(x-10)^2 + (y-5)^2 = (2\sqrt{5})^2$$

$$(6) (a) (i) \frac{6!}{2} = 360$$

$$(ii) [\text{EACH}] NC \therefore \frac{3!}{360} = \frac{6}{360} = \frac{1}{60}$$

$$(iii) 6p_3 = 6 \times 5 \times 4 = 120$$

$$(iv) \text{Case 1: No Cs} \rightarrow HANE = 4p_3 = 24$$

$$\text{Case 2: 1 C} \rightarrow (C) HANE = \left(\frac{4}{1}\right) \times 3! = 36$$

$$\text{Case 3: 2 Cs} \rightarrow (CC) HANE = \left(\frac{4}{1}\right) \times 3 = 12$$

↓

$$(b) T_6 = \left(\frac{11}{5}\right) \left(\frac{2x}{3}\right) \left(\frac{-3}{2x}\right)^5$$

$$= \left(\frac{11}{5}\right) \left(\frac{64x^6}{3^6}\right) \times \left(-\frac{3^5}{32x^5}\right)$$

$$= \left(\frac{11}{5}\right) \left(\frac{2x}{3}\right) (-1)$$

$$= -308x$$

$$\frac{72}{72}$$

$$(c) \text{For } n > 1, LHS = a$$

$$RHS = \frac{1}{2} [2a + 0] = a$$

$\therefore$ True for $n=1$.

Let k be int such that

$$a + (ard) + (ard) + \dots + (a + (k-1)d) = \frac{k}{2} [2a + (k-1)d]$$

Prove $k+1$ is int such that

$$a + (ard) + (ard) + \dots + (a + (k-1)d) + (a + (kd))$$

$$= \frac{k+1}{2} [2a + kd]$$

$$\text{Proof: } LHS = \frac{k}{2} [2a + (k-1)d] + a + kd$$

$$= \frac{k[2a + (k-1)d] + 2a + 2kd}{2}$$

$$= \frac{2ak + k^2d - kd + 2a + 2kd}{2}$$

$$= \frac{2ak + k^2d + 2a + kd}{2}$$

$$= \frac{2a(k+1) + kd(k+1)}{2}$$

$$= \frac{k+1}{2} (2a + kd) = RHS$$

$\therefore$ If true for $n=k$, also true for $n=k+1$

$\therefore$ If true for $n=1$, $\frac{n=1}{n=2}$ $\frac{n=2}{n=3}$ $\frac{n=3}{n=4}$ $\frac{n=4}{n=5}$ $\frac{n=5}{n=6}$ $\frac{n=6}{n=7}$ $\frac{n=7}{n=8}$ $\frac{n=8}{n=9}$ $\frac{n=9}{n=10}$ $\frac{n=10}{n=11}$ $\frac{n=11}{n=12}$ $\frac{n=12}{n=13}$ $\frac{n=13}{n=14}$ $\frac{n=14}{n=15}$ $\frac{n=15}{n=16}$ $\frac{n=16}{n=17}$ $\frac{n=17}{n=18}$ $\frac{n=18}{n=19}$ $\frac{n=19}{n=20}$ $\frac{n=20}{n=21}$ $\frac{n=21}{n=22}$ $\frac{n=22}{n=23}$ $\frac{n=23}{n=24}$ $\frac{n=24}{n=25}$ $\frac{n=25}{n=26}$ $\frac{n=26}{n=27}$ $\frac{n=27}{n=28}$ $\frac{n=28}{n=29}$ $\frac{n=29}{n=30}$ $\frac{n=30}{n=31}$ $\frac{n=31}{n=32}$ $\frac{n=32}{n=33}$ $\frac{n=33}{n=34}$ $\frac{n=34}{n=35}$ $\frac{n=35}{n=36}$ $\frac{n=36}{n=37}$ $\frac{n=37}{n=38}$ $\frac{n=38}{n=39}$ $\frac{n=39}{n=40}$ $\frac{n=40}{n=41}$ $\frac{n=41}{n=42}$ $\frac{n=42}{n=43}$ $\frac{n=43}{n=44}$ $\frac{n=44}{n=45}$ $\frac{n=45}{n=46}$ $\frac{n=46}{n=47}$ $\frac{n=47}{n=48}$ $\frac{n=48}{n=49}$ $\frac{n=49}{n=50}$ 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$$(a) \frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{12x}{1+(3x^2+5)^2} \times \frac{dx}{dt}$$

As $x \rightarrow \infty$,

$$\frac{dy}{dt} \rightarrow 0 \times \frac{dx}{dt} = 0 \times k = 0$$

$$\therefore \frac{dy}{dt} \rightarrow 0$$

$$(b) (i) 0 = t(v \sin \alpha - \frac{1}{2}gt)$$

$$\cancel{t} \neq 0 \text{ or } v \sin \alpha = \frac{gt}{2}$$

$$t = \frac{2v \sin \alpha}{g}$$

$$(ii) x = v \frac{2v \sin \alpha \cos \alpha}{g}$$

$$= \frac{v^2}{g} (2 \sin \alpha \cos \alpha)$$

$$= \frac{v^2}{g} \times \sin 2\alpha$$

$$(iii) t_{\text{max height}} = \frac{v \sin \alpha}{g}$$

$$y = \frac{v \sin \alpha}{g} v \sin \alpha - \frac{g}{2} \frac{v^2 \sin^2 \alpha}{g^2}$$

$$y = \frac{v^2 \sin^2 \alpha}{g} - \frac{v^2 \sin^2 \alpha}{2g} = \frac{2v^2 \sin^2 \alpha}{2g} - \frac{v^2 \sin^2 \alpha}{2g}$$

$$= \frac{v^2 \sin^2 \alpha}{2g}$$

$$(iv) \text{Ratio} = \frac{\frac{v^2 \sin^2 \alpha}{2g}}{\frac{v^2 \sin 2\alpha}{g}}$$

$$= \frac{v^2 g \sin^2 \alpha}{2v^2 g \sin 2\alpha} = \frac{\sin^2 \alpha}{2 \sin 2\alpha}$$

$$= \frac{\sin^2 \alpha}{2 \times 2 \sin \alpha \cos \alpha}$$

$$= \frac{\sin \alpha}{4 \cos \alpha} = \frac{\tan \alpha}{4}$$

$$(v) \text{Max range is } \alpha = \frac{\pi}{4}$$

$$\therefore \text{Ratio} = \frac{\tan \frac{\pi}{4}}{4} = \frac{1}{4}$$

$$(vi) \text{max height} = \text{range}$$

$$\therefore \frac{\tan \alpha}{4} = 1$$

$$\therefore \tan \alpha = 4$$

$$\alpha = \tan^{-1} 4 = 75^\circ 58' \text{ (n.m.)}$$

$$\therefore x = \frac{\sqrt{85} \times 4}{\sqrt{17}}$$

$$x = \sqrt{5} \text{ m}$$

$$y = \frac{\sqrt{85} \times 4}{\sqrt{17}} - 5t^2$$

$$y = 4\sqrt{5} \text{ m} - 5t^2$$

$$y = 4x - 5\left(\frac{x}{\sqrt{5}}\right)^2$$

$$y = 4x - \frac{5x^2}{5}$$

$$y = 4x - x^2$$

$$x^2 - 4x = -y$$

$$x^2 - 4x + 4 = -y + 4$$

$$(x-2)^2 = -(y-4)$$

$$\therefore \text{Vertex } (2, 4)$$

$$\text{Focal length } 4a = 1$$

$$a = \frac{1}{4}$$

$\therefore$ Distance between pole & freeway
is $2 \times \frac{1}{4} = \frac{1}{2} \text{ metre} = 50 \text{ cm}$
 $\therefore$ Yes, can squeeze through
by 10 cm

